Chapter 4 - Sequence and Series

1. Review
2. Harmonic Sequence and Harmonic Mean
3. Properties of Sequence
4. **Proof**: Relation between A.M., G.M., and H.M.
    a. Equation
    b. Inequality
5. Sum of Infinite Geometric Series

#Review

S.n.	Arithmetic Progression (A.P.)	Geometric Progression (G.P.)
1.	**General form:**	
	$a, a + d, a + 2d, ..., a + d(n - 1)$	$a, ar, ar^2, ..., ar^{n-1}$
2.	**Common diff. (d):** $d = t_n - t_{n-1}$ $= t_2 - t_1$ $= t_3 - t_2$	**Common ratio (r):** $r = \frac{t_n}{t_{n-1}}$ $= \frac{t_2}{t_1} = \frac{t_3}{t_2}$
3.	**General term/nth term (t_n),** (n = no. of terms)	
	$t_n = a + d(n - 1)$	$t_n = ar^{n-1}$
4.	**Arithmetic Mean (A.M.)**	**Geometric Mean (G.M.):**
a.	**For 3 terms or 1 mean**	
	$a, m, b \Rightarrow m = \frac{a+b}{2}$	$a, G, b \Rightarrow G = \sqrt{ab}$
b.	**For more than 3 terms or 1 mean,** [n = no. of mean(s)]	
	$a, m_1 m_2, ..., m_n, b$ $d = \frac{b-a}{n+1}$ $m_n = a + nd$	$a, G_1, G_2, ..., G_n, b$ $r = (\frac{b}{a})^{\frac{1}{n+1}}$ $G_n = ar^n$

5.	**Sum of 1st n terms (S_n), (n = no. of terms, b = last term)**	
	$S_n = \frac{n}{2}(a + b)$ $S_n = \frac{n}{2}[a + d(n - 1)]$	$S_n = \frac{br - a}{r - 1}$ $S_n = \frac{a(r^n - 1)}{r - 1}$

#Harmonic Sequence and Harmonic Mean

Harmonic sequence is the sequence obtained by reciprocating the terms of an arithmetic sequence.

Harmonic mean (H.M.) between two numbers a and b is given by H.M. $= \frac{2ab}{a + b}$

#Properties of Sequence

When $k \neq 0$, we have the following properties of sequences:

1. If $a, b, c, d, \ldots$ are in A.S. then
   a. $a \pm k, b \pm k, c \pm k, d \pm k, \ldots$ are in A.S.
   b. $ka,\ kb,\ kc,\ kd, \ldots$ are in A.S.
   c. $\frac{a}{k}, \frac{b}{k}, \frac{c}{k}, \frac{d}{k}, \ldots$ are in A.S.

2. If $a, b, c, d, \ldots$ are in G.S. then
   a. $ka, kb, kc, kd, \ldots$ are in G.S.
   b. $\frac{a}{k}, \frac{b}{k}, \frac{c}{k}, \frac{d}{k}, \ldots$ are in G.S.
   c. $a^k, b^k, c^k, d^k, \ldots$ are in G.S.

   If $a, b, c, d, \ldots$ are in H.S. then
   d. $ka, kb, kc, kd, \ldots$ are in H.S.
   e. $\frac{a}{k}, \frac{b}{k}, \frac{c}{k}, \frac{d}{k}, \ldots$ are in H.S.

#Relation between A.M., G.M., and H.M.

A.M., G.M., and H.M. between two distinct positive numbers holds the following relations:

1. Equation: $(A.M.) \times (H.M.) = (G.M.)^2$
2. Inequality: $A.M. > G.M. > H.M.$

#Sum of Infinite Geometric Series

The Sum of Infinite Geometric Series is given by

$$S_\infty = \frac{a}{1 - r}, (|r| < 1)$$